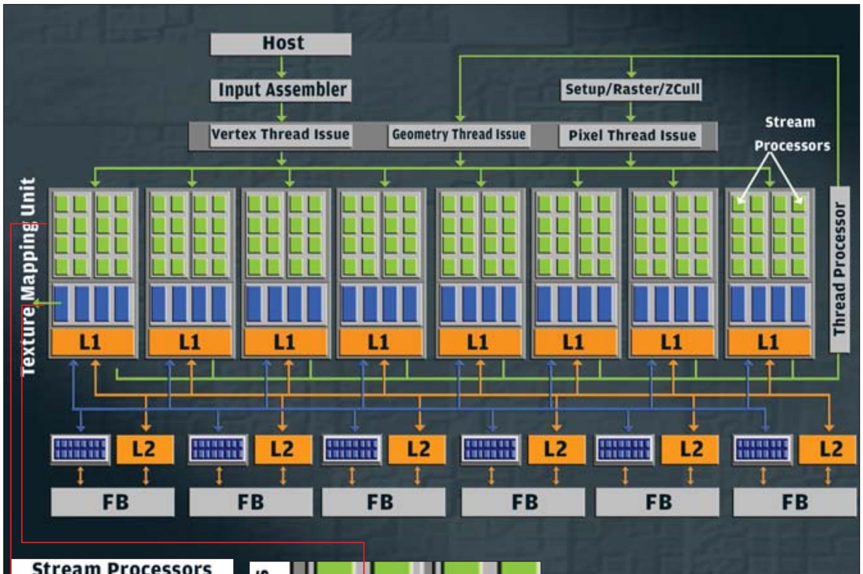

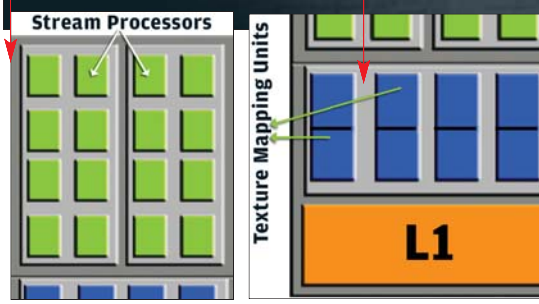

NVIDIA G80's internals: Note the clusters of Stream Processors and Texture Mapping Units

the five threads. The G80 can only handle a single operation (albeit on double the number of threads, that is, 128).

Notice we said "best case" because such a situation is rare in real code, and this peak performance (5x) is not sustainable, and possible only rarely. The performance difference in such a case is restricted to the difference in the number of SPs: 320 / 128 = 2.5x. Do remember that these scenarios are largely code-dependent. In a worst case—where there are no dependent threads—we'd see the G80 with a 2x advantage, that is, 128 threads vs. 64. All the above are hypothetical situations, and we'll see developers optimising code for one or the other GPU.

There's another variable to consider, in that the G80's shaders are clocked at much higher speeds (see table *Matters Of The Core*). In fact, at double the speed of the R600 (1500 MHz vs. 740 MHz), the overall throughput should remain the same. This is more or less confirmed when both companies reveal their GPUs' performance figures—475 GigaFLOPS on the R600, and 520 GigaFLOPS on the G80.

In our opinion, in practice, therefore, the G80 should in most cases be significantly faster than the R600, unless code has been heavily optimised for the R600, wherein its heavy-duty computing engine comes into use.

Texturing: How Much Is Enough?

There are two methods in vogue for displaying spectacular visual effects in games. The older method employs complex textures specifically, consisting of complex code written for each texture, while newer games—in general; there are exceptions—use shaders. All GPUs now feature programmable hardware shaders, for which code can easily be developed.

Look at the table *Matters Of The Core* and you'll see the Texture Mapping Units (TMUs) on the R600 remain unchanged from the earlier Radeons at 16 (texture filter units). The TMU on R600 is brand-new, with many improvements over predecessors, namely improved Anisotropic Filtering, filtering of FP 32-bit textures, in addition to support for larger texture samples in compliance with DX 10.

The TMUs on the R600 are additionally able to work on both unfiltered and filtered textures. While filtered textures mostly represent image-based data like pixel textures, unfiltered textures represent vertex data and other blocks of data (which are not related to images). In true compliance with DX 10, the R600 can display FP formats from 11:11:10, (R:G:B), up to 128-bit.

NVIDIA's G80 is at a direct advantage with double the number of TMUs (32). Although NVIDIA has double the number of texture filtering units (64), the number of texture address units is only 32. The G80 only works on filtered texture samples, somewhat limiting its capabilities (keeping in mind true unified architecture), but texture filtering should be faster on the G80 because of sheer numbers.

Pixel Mascara: Shading Up To Kill

Pixel operations bring up the last few steps in the rendering pipeline before the output is diverted to the screen. ATI's nomenclature for this part of the core is *Render Back Ends*, while NVIDIA uses the handle *ROP Units* (Raster Operation Units). The ROP unit is basically a complex processor responsible for compression and decompression of textures, Alpha Blending, frame buffer, and Anti-Aliasing, to name a few functions.

The R600 has the same number of ROPs as its predecessor, that is, four; one for each SIMD unit. The maximum throughput these four ROPs maintain is 16 pixels per clock (four pixels per ROP). Although ATI has mentioned improvements in efficiency over the previous-generation R580+, the huge number of shader units (320 or 64, whichever way you look at it) will be seriously bottlenecked in the ROP portion of the R600's hardware pipeline.

The G80 has six ROP units, each capable of four operations per clock. That's 24 Raster Operations per clock on the whole. So while theoretically the G80 has more processing power, remember here that the ROPs are clocked at core speeds, so the R600 should be able to hold its own here (740 MHz vs. 648 MHz), with maybe a slight edge going to NVIDIA's hardware. We don't expect the difference to effectively more than 10 per cent either way.

Memory Bandwidth: Multi-Lane Traffic Control!

ATI's *Ring Bus* was talked about on the R500 series. This was basically dual, 256-bit unidirectional buses used to carry blocks of vertex, pixel, and data around the various subsystems of the core